

AHMER KAMRAN

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PROFESSIONAL SUMMARY

Computer Engineering Graduate with practical experience in Digital IC Design, Verilog HDL, FPGA Development, Physical Design (OpenLane/OpenROAD), Embedded Systems, and IoT. Skilled in RTL design, synthesis, FPGA implementation, and hardware-software integration. Experienced in developing real-time embedded systems, computer vision applications, and digital design projects through academic and practical work.

EDUCATION

UIT University — BE Electrical Engineering (Computer Systems)

July 2026

Karachi, Pakistan

CGPA: 2.92

WORK AND PRACTICAL EXPERIENCE

IC Design Trainee| Chip Select

May 2026 – Present

- Developed and implemented RTL designs in Verilog HDL, including digital logic, FSMs, counters, ALU, FIFO, SRAM, and pipelined multiplier.
- Performed FPGA implementation and verification of RTL designs; designed UART and VGA controllers.
- Performed RTL-to-GDSII physical design using LibreLane/OpenLane/OpenROAD, including synthesis, STA, floorplanning, placement, CTS, routing, and GDSII generation.
- Analyzed area, cell count, timing, and implementation metrics for SPM and Comparator designs.

Excelerate

Feb 2026 – Mar 2026 (Remote)

- Conducted exploratory data analysis (EDA) and visual trend modeling using Python and AI tools.
- Prepared structured evaluation reports and data dashboards highlighting summary statistics and predictive insights.

Data Operation Intern| Pixel Canva

Feb 2026 – Mar 2026 (Remote)

- Managed data pipeline operations, structured organizational data tracking, and maintained high data accuracy.
- Administered targeted email distribution campaigns, handling communication workflows and administrative tracking logs.

IT Support Intern

2025

- Maintained local workstation hardware components and performed routine system diagnostics.
- Provided technical troubleshooting and user support for network and software infrastructure issues.

TEACHING AND VOLUNTEER EXPERIENCE

Electronic & Technology Workshop Instructor| Korangi Academy(NGO)

2025

- Taught hands-on electronics, circuit simulation, and hardware prototyping to school students using Tinkercad and basic microcontrollers.
- Mentored young learners on foundational engineering concepts, digital logic, and sensor integration.

Instructor| Logic Unlocked

2025

- Ran sessions on logical thinking, basic computing, and problem-solving for student groups.

Event Volunteer| Iqra University

2024

(Volunteer) Robotics Competition

- Helped organize and run a university-level robotics competition with technical teams.

PROJECTS

Smart Swarm Energy Distribution and Monitoring System

ESP32 | Arduino IDE | ESP-NOW | WiFi | IoT

- Designed an IoT-based decentralized smart energy distribution system.
- Implemented wireless communication between multiple energy nodes using ESP-NOW.
- Monitored voltage, current, and power consumption in real time.
- Developed smart energy sharing and load management mechanisms.
- Integrated a real-time monitoring dashboard for energy control.

UART Controller Design Using Verilog

Verilog | FPGA

- Designed UART transmitter and receiver modules using Verilog HDL.
- Verified functionality through simulation and waveform analysis.
- Synthesized RTL design for FPGA implementation.

FPGA-Based VGA Controller Using Verilog

Verilog | FPGA

- Designed a VGA controller to generate synchronization signals.
- Implemented graphical output display on FPGA hardware.
- Verified display timing and hardware functionality.

ASIC Physical Design and Layout Generation

OpenLane | OpenROAD | Sky130 | Ubuntu Linux

- Implemented RTL-to-GDSII flow for SPM and Digital Comparator designs.
- Performed synthesis, floorplanning, IO placement, and placement using OpenLane/OpenROAD.
- Analyzed area, timing, utilization, and power reports.
- Generated physical layouts using open-source ASIC design tools.

TECHNICAL SKILLS

- **Programming Languages:** Python, Verilog, C, C++
- **Digital Design:** RTL Design, FSM Design, UART, VGA Controller, FPGA Design, Digital Logic Design, Timing Analysis
- **ASIC Physical Design:** OpenLane, OpenROAD, Sky130 PDK, Synthesis, Floorplanning, IO Placement, Placement, Layout Generation, Timing Reports
- **IoT & Embedded Systems:** ESP32, Arduino IDE, Raspberry Pi, ACS712, MOSFET Switching, Logic ICs (7400 Series), GPIO Programming
- **AI, ML & Data Science:** Natural Language Processing (NLP), TF-IDF, Cosine Similarity, Computer Vision (OpenCV), Pandas, NLTK, Scikit-learn
- **Software & Tools:** Tinkercad, Logisim, Proteus, MATLAB, AWS (EC2, S3), MS Project, Primavera P6

CERTIFICATIONS & COURSES

- **Cloud Computing Course – NED Academy (PITP):** Completed practical training in cloud computing concepts and technologies through the Pakistan IT Industry Program (PITP).
- **Certificate of Appreciation – Event Coordinator, TechFest '24:** Recognized for leading organizational operations and coordinating event activities.
- **Certificate of Participation – TechFest '24, Iqra University:** Participated in TechFest '24 and contributed to the technical event activities.
- **Certificate of Appreciation – Tinkercad & Electronics Summer Camp 2025, Korangi Academy:** Recognized for contribution to hands-on electronics and Tinkercad learning activities.
- **Digital Freelancing Course – UIT Freelancers Club:** Completed a digital freelancing course organized by UIT Freelancers Club and conducted by Vorker at UIT University.